北京交通大學

信号与系统

School of Electronic
and Information Engineering

Zhou Hang
2026.3

Assessment requirements

- Final score breakdown
  - Midterm 20%
  - Final 50%
  - Assignments (Submission once， No late) and Participation 10%
  - Labs 15%
  - On-line studies(http://www.icourse163.org/course/NJTU-359003) 5%

  (Register your nickname as the format”bjtuwhss19726077赵祥文”)
- Assignments
  - Will be given after each class
  - Collection time: Each week’s Mon (week 2-11)
  - Upload the corrected homework each time and lab report to zh_xhyxt2021@126.com with bjtu email.

  If any changed later, the lectuer will inform…

Lecturer and teaching assistant introduction

Lecturer： **周航 Zhou Hang**

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刘熙喆 23721048@bjtu.edu.cn 2402

Representative： Recruiting…

Requirement

To do:

1. The simulation experiment is divided into groups of 6-7 people, and the group leader will be elected.

2. Class representative should have 2 people to sign up.

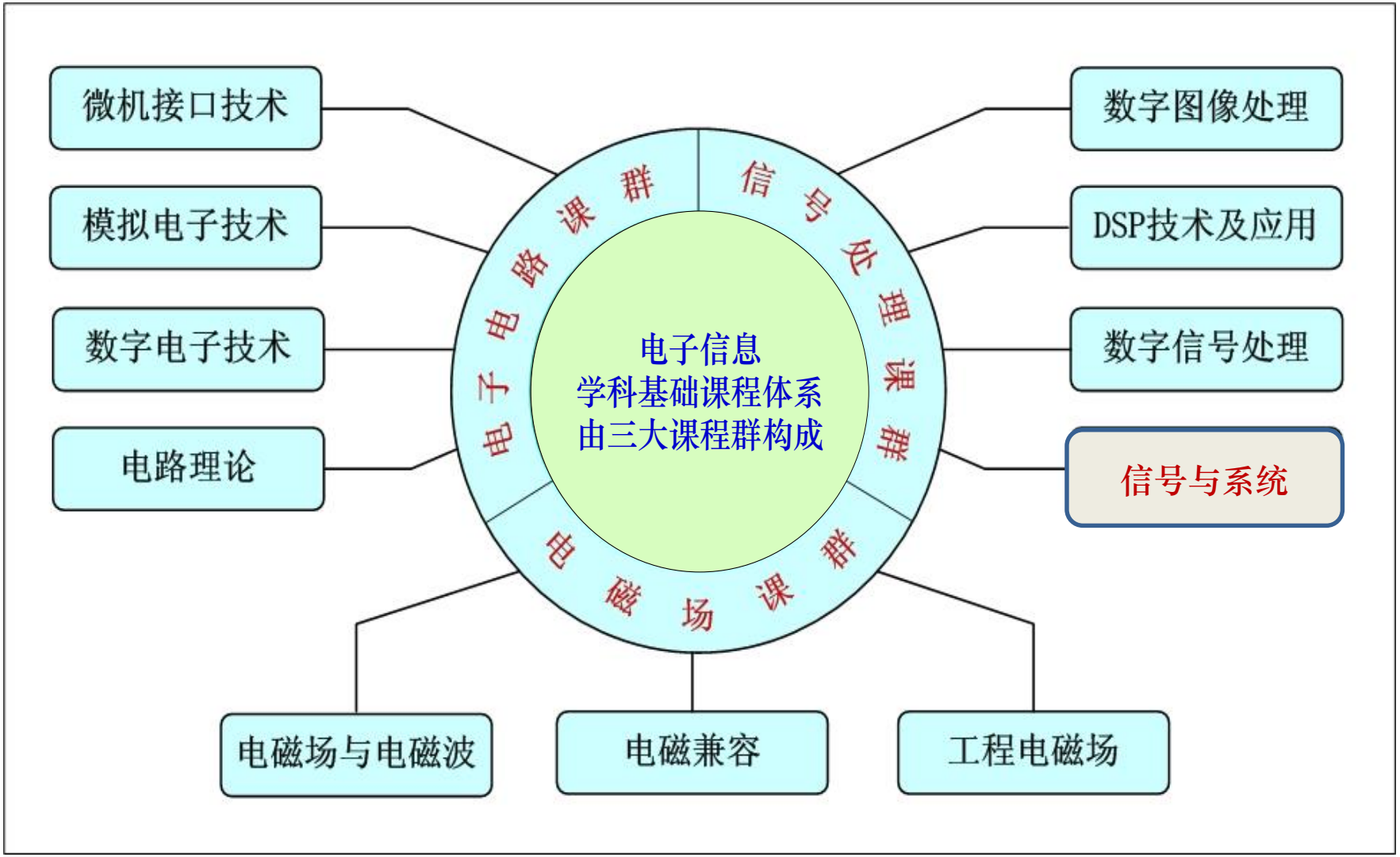
Chapter 1 Introduction to Signal and System Analysis

- Introduction
- Signal description and classification
- System description and classification
- Applications of signals and systems

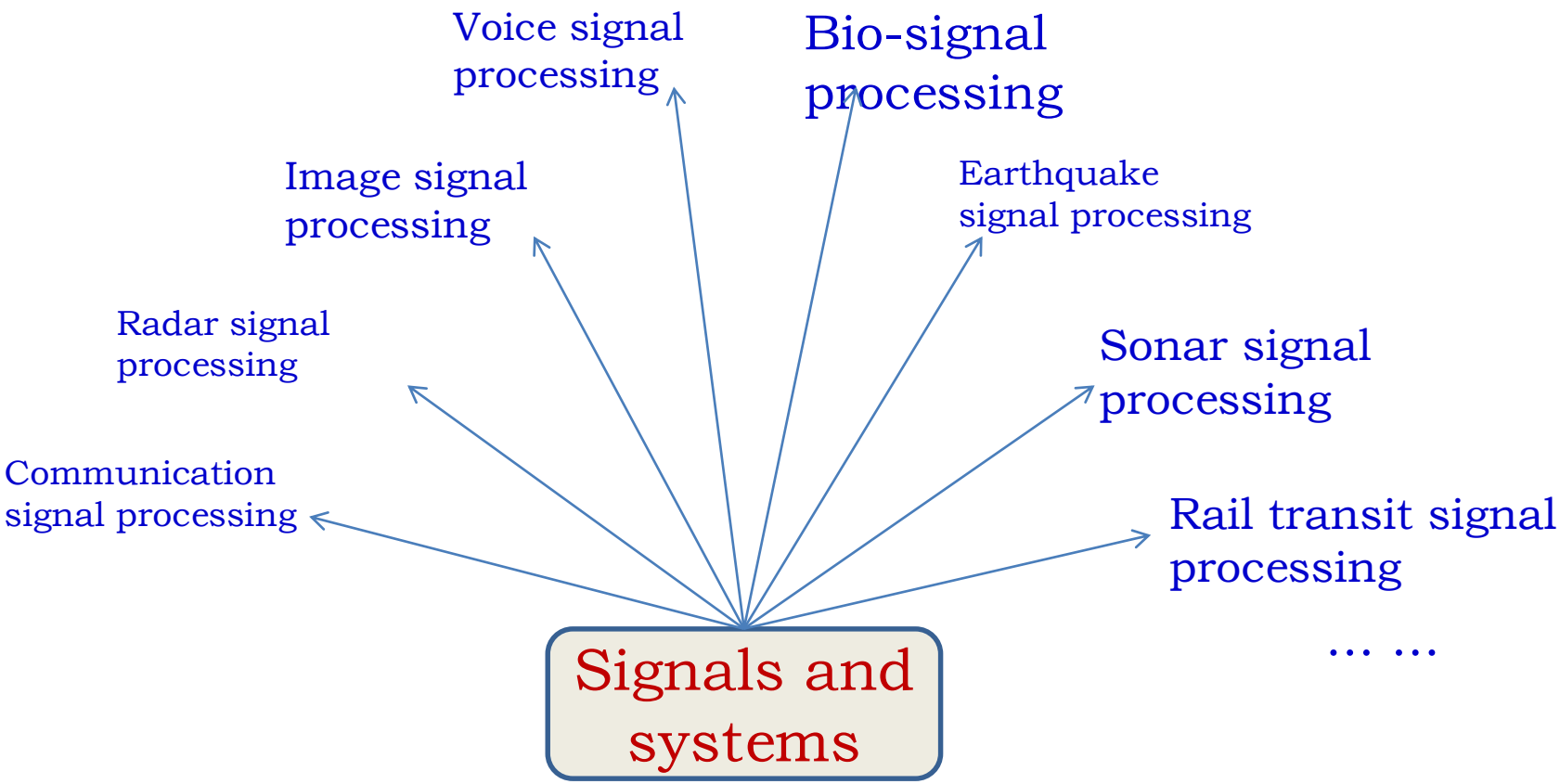
Introduction

- Electronic information professional course system
- Signals and systems course content
- Application Examples of Signal and System Theory
- Signals and Systems Course Teaching Objectives
- Signals and Systems Course Teaching Resources
- Teaching philosophy of contemporary engineering education

1. Electronic information professional course system



1. Electronic information professional course system



2.Basic content

Course connotation :

Signal representation
System description

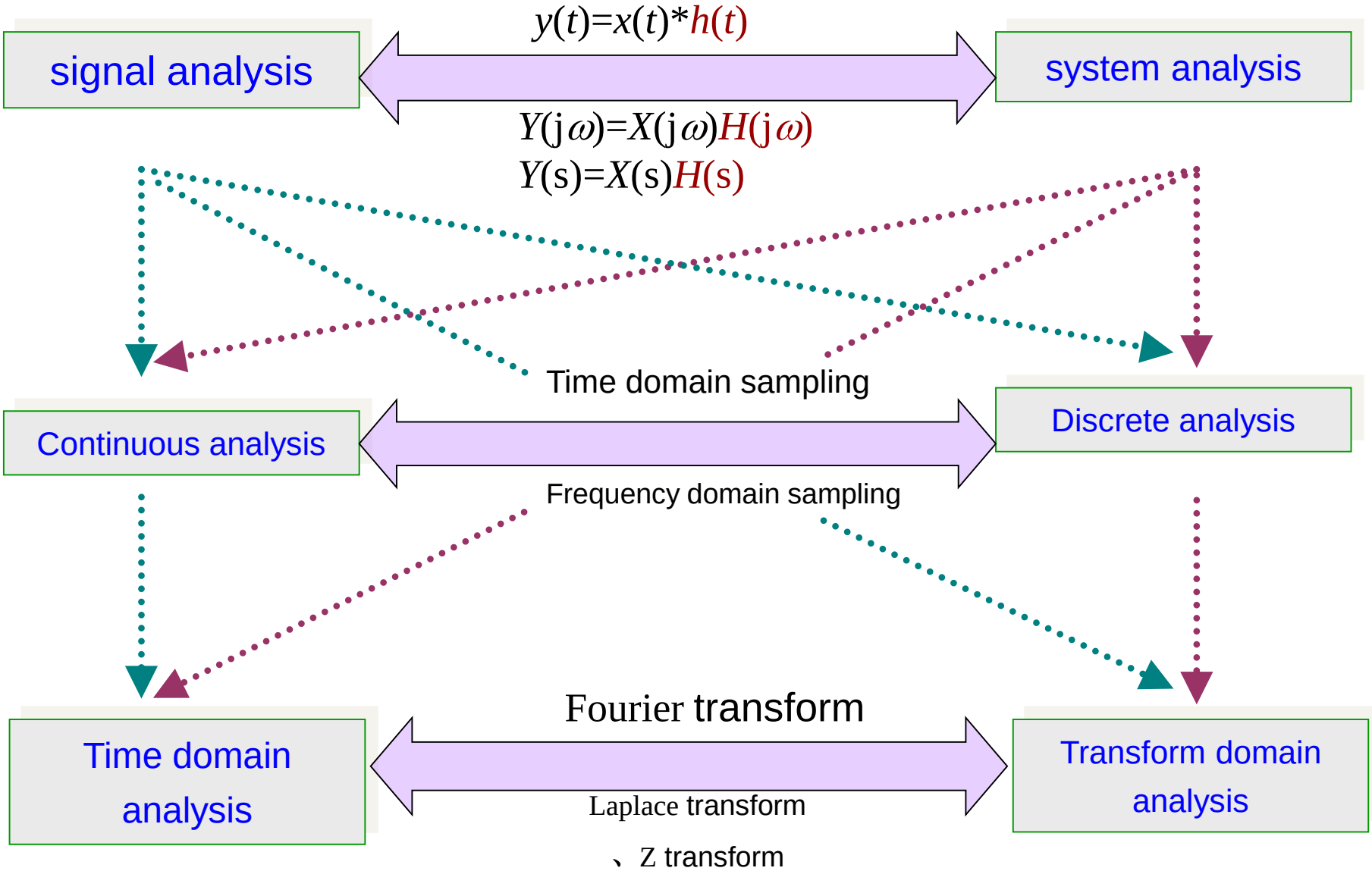
2.Basic content

- 信号与系统
  - 信号分析
    - 连续信号
      - 时域表示：信号表示为冲激信号的线性组合
      - 频域表示：信号表示为正弦信号的线性组合（CFS，CTFT）
      - S域表示：信号表示为复指数的线性组合（单边、双边）
    - 抽样 ⇕ 重建
    - 离散信号
      - 时域表示：信号表示为脉冲序列的线性组合
      - 频域表示：信号表示为正弦序列的线性组合（DFS，DTFT）
      - Z域表示：信号表示为复指数的线性组合（单边、双边）
  - 系统分析
    - 连续系统
      - 时域描述：微分方程(组)：$h(t)$　$y(t)=x(t)*h(t)$
      - 频域描述：系统频响特性：$H(j\omega)$　$Y(j\omega)=X(j\omega)H(j\omega)$
      - S域描述：系统函数：$H(s)$　$Y(s)=X(s)\,H(s)$
    - 离散系统
      - 时域描述：差分方程(组)：$h[k]$　$y[k]=x[k]*h[k]$
      - 频域描述：系统频响特性：$H(e^{j\Omega})$　$Y(e^{j\Omega})=X(e^{j\Omega})H(e^{j\Omega})$
      - Z域描述：系统函数：$H(z)$　$Y(z)=X(z)H(z)$

Signal representation

System description

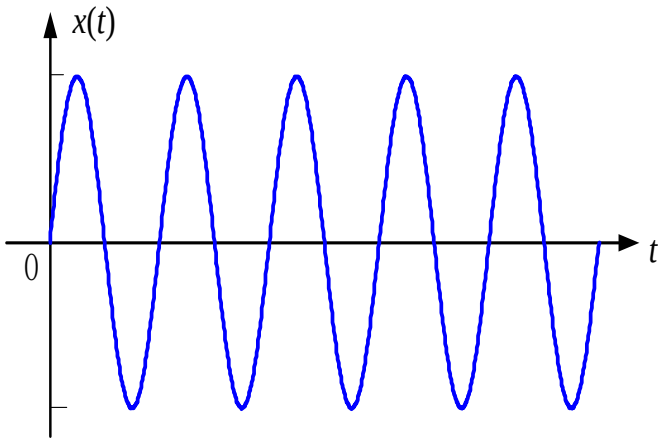
2.Basic content



2.Basic content

- The signal must be generated, sent, transmitted and received by the system, and there is no isolated signal from the system。

Signal generator



Signals and systems are interdependent

2.Basic content

- Signals must be generated, sent, transmitted, and received by the system, and there is no signal that exists in isolation from the system.

voice
send image
word

voice
receive image
word

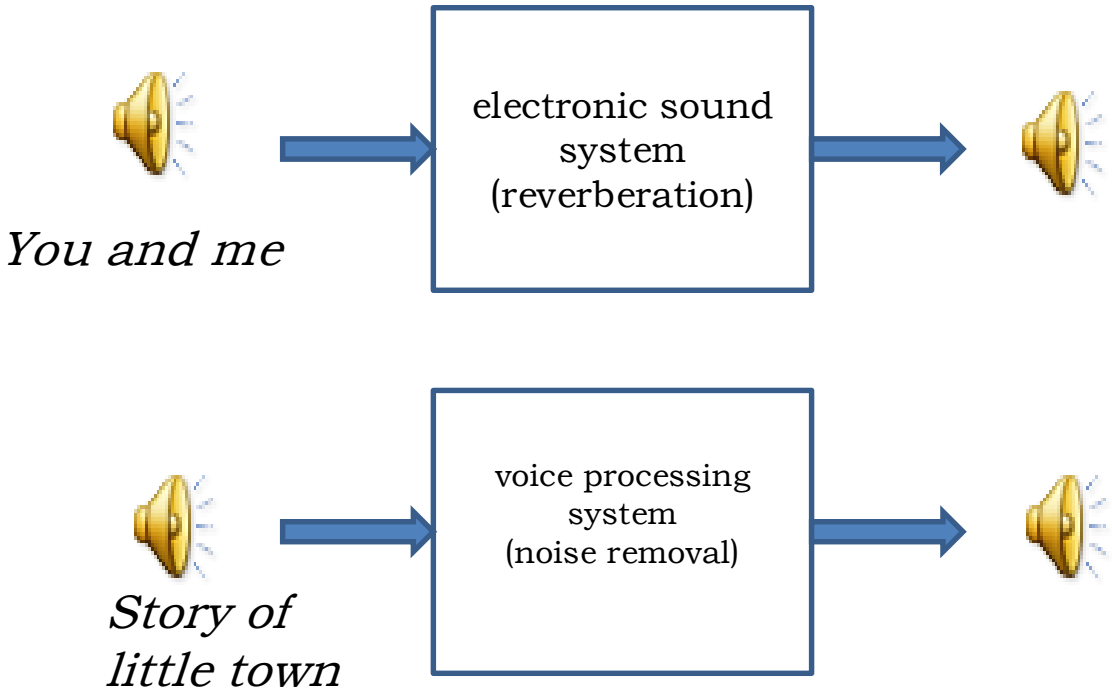
Signals and systems are interdependent.

“Godzilla”, 2014

Bury me E

2.Basic content

- The important function of the system is to process, transform and process the signal, and the system without signal processing has no meaning of existence.



Signals and systems are interdependent.

3. Application Examples of Signal and System Theory

- Call, sending text messages, Weibo, Wechat

voice image
word
transmission

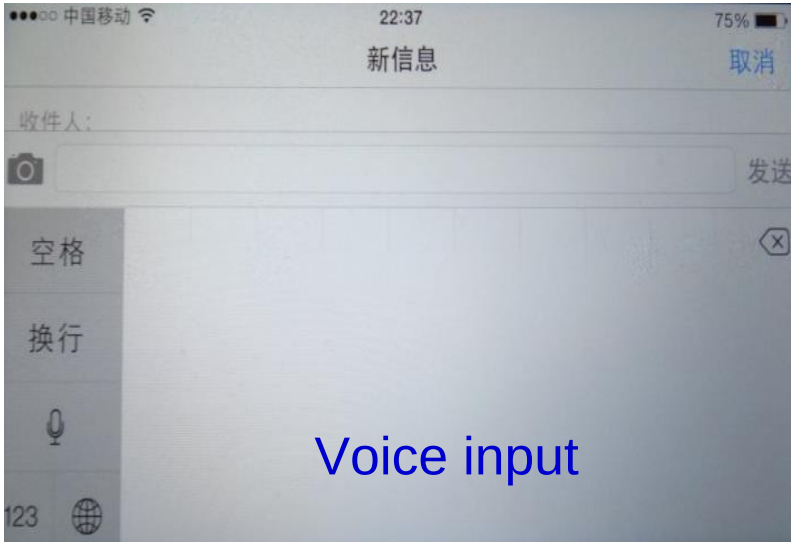
While calling, sending text messages, Weibo, and Wechat, information transmission has also started.

Knowledge module

- Mobile communication
- Information theory
- Communication principle
- stochastic process
- digital signal processing
- Signals and Systems

3. Application Examples of Signal and System Theory

- Voice input、sound email



Corresponding knowledge module

Speech Signal Processing

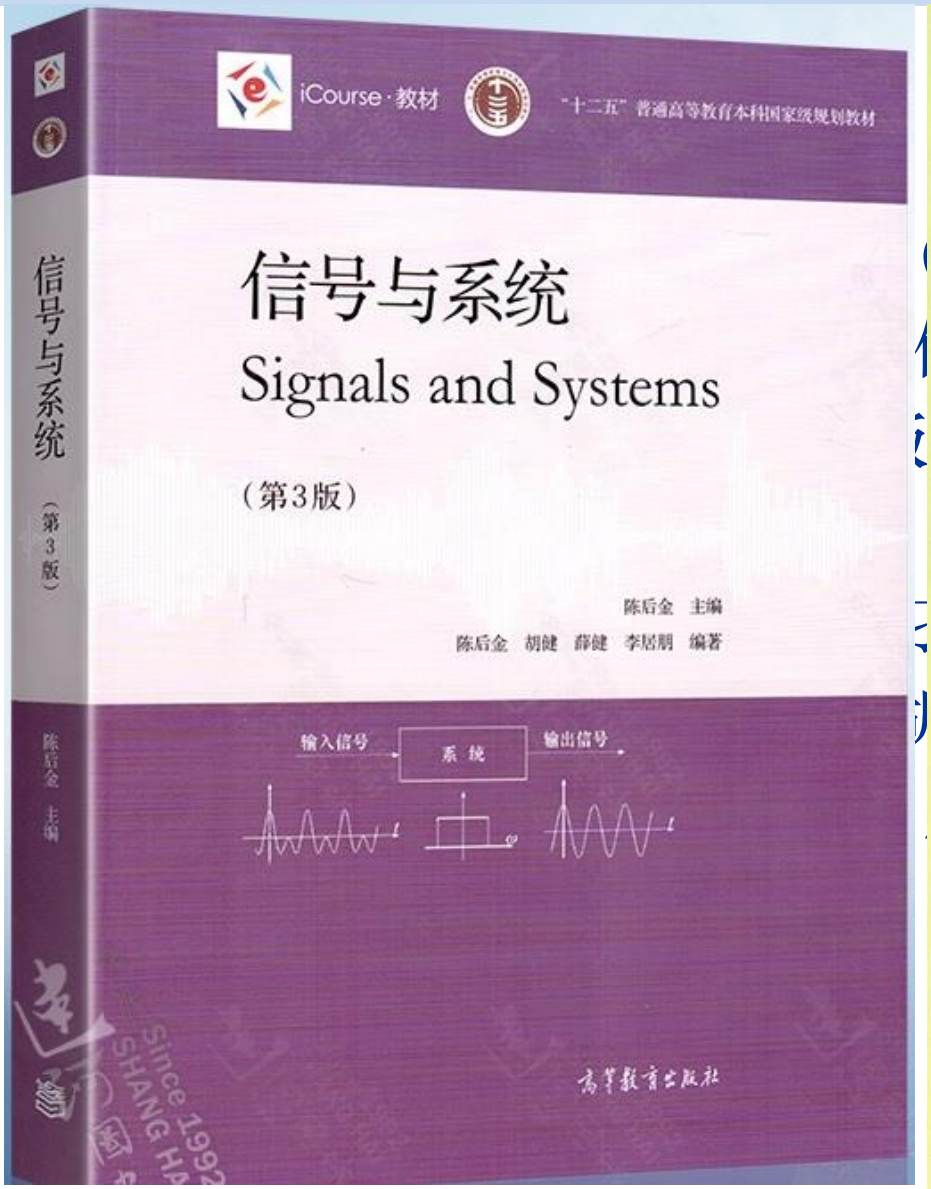
Digital signal processing

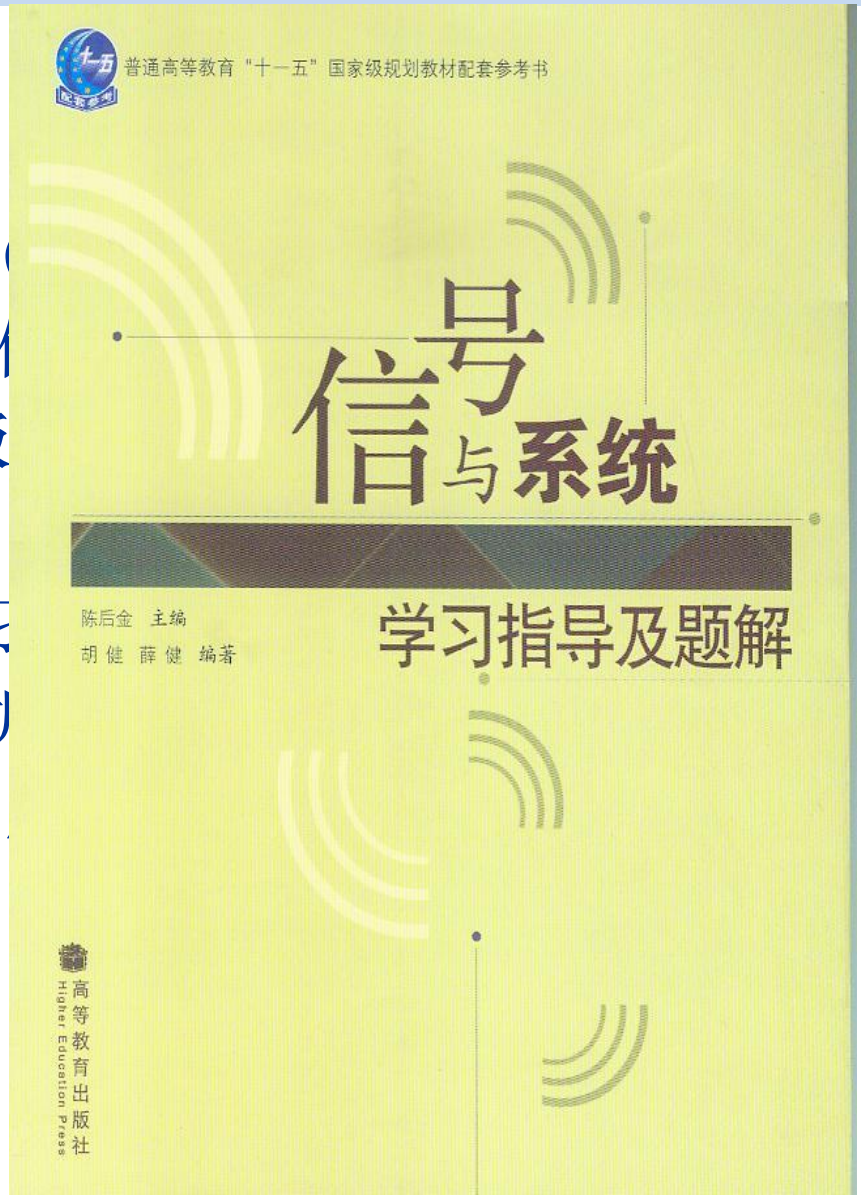
Signals and Systems

4. Signals and Systems Course Teaching Objectives

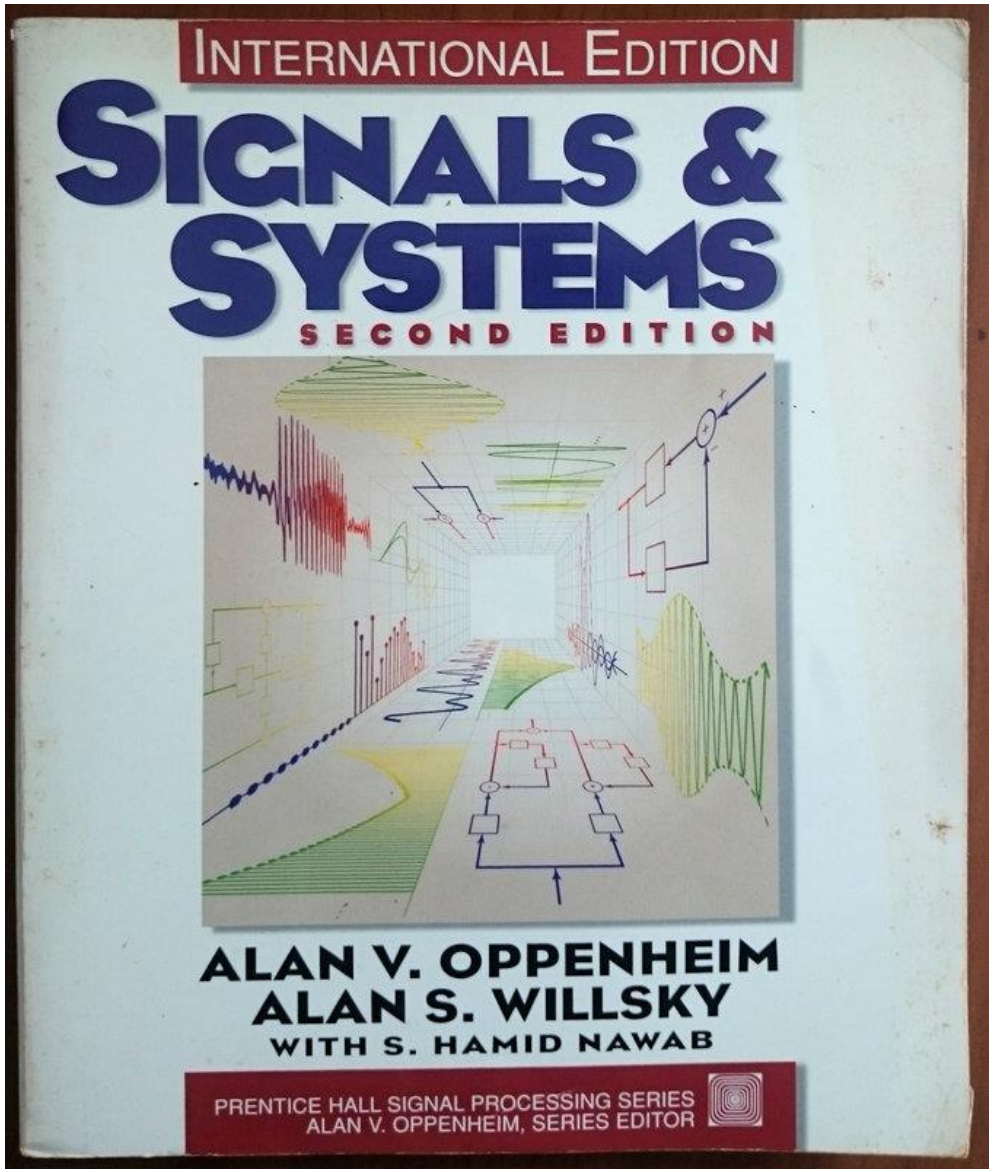
- ※ Firmly grasp the basic principles and basic analysis methods of signals and systems;
- ※ Master the time domain and transform domain analysis methods of signals and systems, and deeply understand the mathematical concepts, physical concepts and engineering concepts of various signal transforms (Fourier transform, Laplace transform, z-transform);
- ※ Master the basic ideas of signal representation and system description, and lay a solid theoretical foundation for further study of subsequent courses;
- ※ Exercise students' analytical and problem-solving abilities, as well as their autonomous learning and inquiry abilities.

5. Signals and Systems Course Teaching Resources





5. Signals and Systems Course Teaching Resources



5. Signals and Systems Course Teaching Resources

New version Signals and Systems

https://www.xuetangx.com/course/bjtu08071005219intl/8293043

6.Teaching philosophy of contemporary engineering education

University engineering education can't just stop at letting students "know", but not guide students to internalize "knowledge" into "knowledge", and then put it into "action". Otherwise, students will only know some definitions and properties in books, and exercise calculus.

If students are only satisfied with memorizing the contents of these books, it will be difficult to obtain true knowledge and insights, let alone innovation. Students should think carefully on the basis of existing knowledge, understand and apply what they have learned through reading between the lines, so as to form their own knowledge and ability, and become excellent talents who "knowledge well, learn well and use it well".

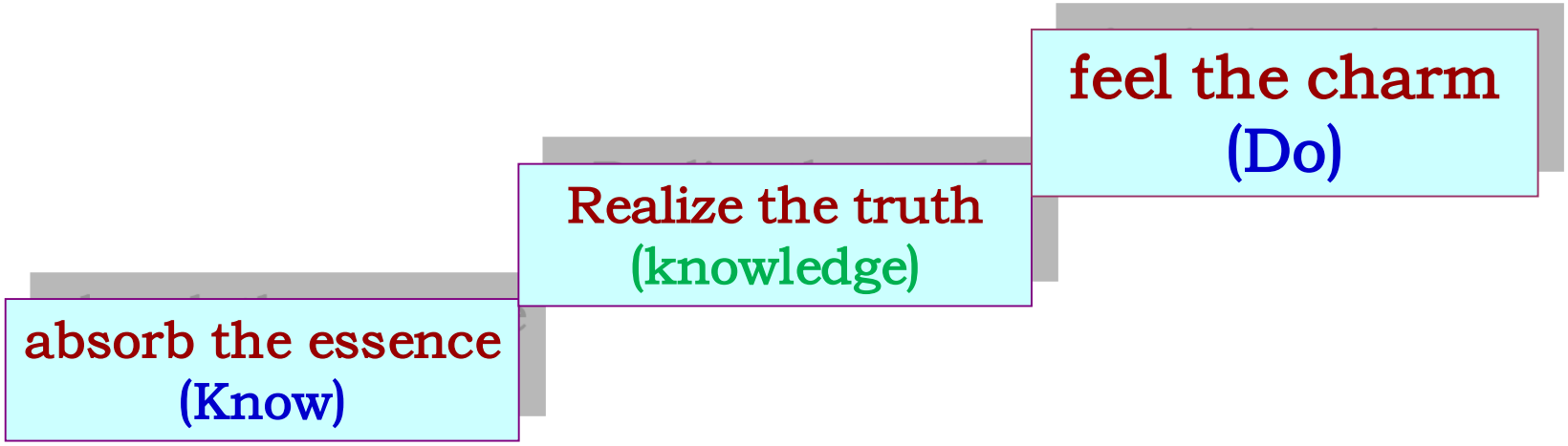
6.Teaching philosophy of contemporary engineering education

University education should gradually change from focusing on the teaching of book content to using teaching content as a carrier to inspire and guide students.

Absorb the essence of knowledge in learning;

Comprehend the true meaning of knowledge in inquiry;

Feel the charm of knowledge in practice.



6.Teaching philosophy of contemporary engineering education

The focus of teaching should be to clarify "why", "what" and "what to do". Abstract definitions should be made from mathematical concepts, properties should be explained from physical concepts, and applications should be interpreted from engineering concepts, so as to stimulate students' interest in learning and enhance students' inquiry learning ability.

Education is what remains after one has forgotten everything he learned in school.

Universities should give students more knowledge edification, quality cultivation, and ability training.

Thanks

Some of the materials quoted in this course are accumulated by the main lecturer over the years, from a variety of media and colleagues, peers and friends of the exchange, it is difficult to attribute them all, so I would like to state and thank you.